

From Hive Minds to Administered Minds: A Formal Theory of Agency Collapse and Emergent Macro-Agency

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Abstract

In Lerer (2025a), I argued that game theory's foundational assumptions formalize the structural conditions of human agency. Here I formalize this intuition by establishing an equivalence with hive mind dynamics. Drawing on Soma et al.'s (2025) proof that populations following imitation rules are mathematically equivalent to single reinforcement learning agents, I show that the four conditions of strategic agency (separation, imperfect information, bounded rationality, minimal symmetry) map onto parameters governing the transition from distributed cognition to unified macro-agency.

The central insight is counterintuitive: when agency conditions erode, agency does not disappear. It transfers to a higher level of organization. The administered population becomes a macro-agent optimizing objectives potentially orthogonal to individual welfare.

I develop a Hive Mind Index measuring proximity to agency collapse and apply it to authoritarian systems, social media platforms, and AI-human interaction. The framework connects to constitutional vocabulary: in Lerer (2025b), I documented that "general welfare" and "common good," though philosophically synonymous, evolved a 17.8-fold fitness differential, with "common good" achieving maximum fitness exclusively during authoritarian periods. The Hive Mind framework explains this: "common good" functions as a homogenizing reward signal attacking preference heterogeneity, while "general welfare" preserves it. Platform vocabulary ("community standards") functions analogously.

Keywords: agency collapse, hive mind, reinforcement learning, swarm intelligence, game theory, extended phenotype, institutional design, constitutional vocabulary

JEL Codes: D82, D83, K00, P16, B53

I. Introduction: The Missing Piece

In a companion paper, I argued that game theory's foundational assumptions formalize the structural conditions of human agency (Lerer, 2025a). When these conditions erode, game theory fails to describe observed interactions. This failure is diagnostic: it signals that something has changed in the environment, not that the models need refinement. The four conditions, separation of players, imperfect information, bounded rationality, and minimal symmetry, correspond to requirements for genuine strategic agency. When they collapse, interaction transforms from game to administration.

That paper left a question unanswered. When agency erodes, what happens to it? The implicit assumption was that agency simply diminishes, that administered subjects have less of something they previously possessed. But recent work in swarm intelligence suggests a more

disturbing possibility: agency may not diminish but transfer. The individual loses strategic capacity, but a macro-entity gains it.

Soma et al. (2025) proved that populations of organisms following simple imitation rules are mathematically equivalent to single reinforcement learning agents. A swarm of honey bees, each following local rules about whom to imitate, collectively constitutes a learning system that optimizes nest site selection. The individual bee is not an agent in any meaningful sense; it follows mechanical rules. But the swarm is an agent: it learns, adapts, and converges on optimal choices through a process formally identical to bandit algorithms used in machine learning.

This paper establishes the connection between these two frameworks. The erosion of agency conditions described in Lerer (2025a) corresponds precisely to the emergence of hive mind dynamics described in Soma et al. (2025). When individuals lose the capacity for strategic action, they do not become inert. They become components of a higher-level agent whose objectives may diverge radically from their own.

The implications extend beyond theoretical interest. Understanding how macro-agency emerges from individual agency collapse clarifies what is at stake in debates about surveillance, propaganda, platform monopolies, and artificial intelligence. The question is not merely whether individuals retain freedom but whether the collective entities they constitute serve purposes aligned with human flourishing.

II. Two Frameworks, One Phenomenon

II.A. The Canary Framework: Four Conditions of Agency

In Lerer (2025a), I identified four assumptions of game theory that correspond to structural conditions for strategic agency:

Condition 1 (Separation): Game theory assumes multiple players with non-identical preference orderings. This formalizes the existence of distinct individuals with private spheres and diverse ends. When separation collapses, through ideological homogenization, absorption into collective identities, or elimination of private preferences, the "game" reduces to a coordination exercise with a single effective preference.

Condition 2 (Imperfect Information): Games of incomplete information assume that players are uncertain about each other's types, constraints, and private knowledge. This uncertainty creates the space for strategy. When one party can predict another's behavior with high accuracy, the predicted party ceases to be an agent and becomes a variable.

Condition 3 (Bounded Rationality): Human players cannot compute optimal strategies; they satisfice, use heuristics, make errors. This unpredictability creates space for strategic innovation. When cognition becomes uniform and predictable, whether through propaganda, algorithmic management, or technological intervention, genuine choice disappears.

Condition 4 (Minimal Symmetry): Even asymmetric games require that each player can affect outcomes. When power differentials become extreme, the weaker party's "choices" reduce to compliance or destruction. They are not playing; they are being administered.

The Canary framework provides a diagnostic: when game-theoretic models fail, check which conditions have eroded. But it does not explain what replaces strategic interaction when conditions collapse.

II.B. The Hive Mind Framework: Collective Intelligence as Single Agent

Soma et al. (2025) established a mathematical equivalence between collective decision-making through imitation and individual reinforcement learning. Their key results:

Proposition 1: A population following "imitation of success" rules (where individuals copy more successful neighbors) aggregates to the Taylor Replicator Dynamic, which is equivalent to the Cross Learning update rule in reinforcement learning.

Proposition 2: A population following the "weighted voter model" (where individuals imitate neighbors at rates proportional to those neighbors' success signals) aggregates to the Maynard-Smith Replicator Dynamic, equivalent to what Soma et al. call Maynard-Cross Learning.

In formal terms, for a population with types $a \in \{1, \dots, n\}$ and population vector $\pi = (\pi_1, \dots, \pi_n)$:

The weighted voter model yields:

$$d\pi_a = (\pi_a/v^p)(q_a^p - v^p)$$

where q_a^p is the expected payoff of type a and v^p is the average population payoff. This is mathematically identical to the expected update of a single reinforcement learning agent using Maynard-Cross Learning.

The critical insight: individual bees are not learning. They follow blind imitation rules. But the swarm learns. The population vector π functions simultaneously as a distribution of individual types and as the policy of a macro-agent. Each individual is an action sample; the swarm is the agent.

II.C. The Synthesis: Agency Transfer, Not Agency Loss

The connection between these frameworks is precise. Consider what happens when each agency condition collapses:

Separation collapse → Type homogenization: When individuals lose distinct preferences, they converge toward shared types. In the Soma et al. framework, this corresponds to the population vector π converging toward a deterministic distribution. The "game" among individuals disappears, but the macro-agent's policy becomes more coherent.

Information collapse → Imitation dominance: When individuals cannot maintain private information, they must rely on observable signals from others. This is precisely the weighted voter model: individuals imitate based on perceived success signals. The more complete the information collapse, the more closely the population follows replicator dynamics.

Bounded rationality collapse → Rule-following uniformity: When cognitive diversity disappears, individuals follow identical decision rules. This is the limiting assumption of evolutionary game theory: individuals do not strategize but follow fixed protocols. The macro-agent emerges precisely because individuals are not thinking.

Symmetry collapse → Environmental parallelization: When individuals cannot affect macro-outcomes, they become parallel samples of the same environment. Each individual is a bandit arm being pulled; the macro-agent learns from the aggregate.

The synthesis is captured in a single statement:

When the four conditions of individual agency collapse, the population transitions from a system of interacting strategic agents to a single macro-agent learning through reinforcement.

This is not metaphor. It is mathematical identity. The same equations describe both the collapse of individual strategic capacity and the emergence of collective learning.

III. Formal Equivalence

III.A. Mapping Conditions to State Variables

The four agency conditions can be operationalized as measurable parameters:

Condition	Agency Parameter	Hive Mind Parameter	Collapse Means
Separation	Preference heterogeneity: $\text{Var}(U_i)$	Type distribution: π	$\text{Var}(U_i) \rightarrow 0$; $\pi \rightarrow \delta_k$ (single type)
Information	Private signal independence: $I(s_i; s_j)$	Imitation rate: β	$I \rightarrow 0$; $\beta \rightarrow 1$ (full imitation)
Bounded Rationality	Policy diversity: $\text{Var}(\pi_i)$	Update uniformity	$\text{Var}(\pi_i) \rightarrow 0$; all follow same rule
Symmetry	Individual impact: $\partial V / \partial a_i$	Learning rate $\partial V / \partial a_i \rightarrow 0$; individual contribution: $\alpha = 1/N$	negligible

The collapse trajectory can be formalized. Let $\theta \in [0, 1]$ represent the degree of agency collapse, with $\theta = 0$ indicating full individual agency and $\theta = 1$ indicating complete hive mind dynamics.

For each condition $c \in \{\text{separation, information, rationality, symmetry}\}$, define:

- θ_c = degree of collapse of condition c

The aggregate collapse parameter:

$$\Theta = f(\theta_{\text{separation}}, \theta_{\text{information}}, \theta_{\text{rationality}}, \theta_{\text{symmetry}})$$

where f is a function capturing the interaction effects. A simple specification:

$$\Theta = 1 - \prod^c (1 - \theta_c)$$

This captures the intuition that collapse of any single condition moves the system toward hive mind dynamics, with multiplicative interaction.

III.B. The Transition Dynamics

The transition from distributed agency to macro-agency follows identifiable dynamics.

Phase 1: Distributed Agency ($\Theta \approx 0$)

Each individual i maximizes their own value function:

$$V_i = E[\sum^t \gamma^t r_i(t) \mid s_i, a_i]$$

Interaction is genuinely strategic. Game-theoretic models describe and predict. The population is not a coherent agent; it is a system of interacting agents.

Phase 2: Transition ($0 < \Theta < 1$)

As conditions erode, individual optimization becomes increasingly constrained. Imitation dynamics emerge:

$$\pi_i(t+1) = (1-\beta)\pi_i(t) + \beta \cdot \text{imitate}(\pi_{\text{successful}})$$

where β increases with information collapse and preference homogenization. Individual policies converge. Strategic diversity decreases. The population vector π begins to function as a coherent policy.

Phase 3: Hive Mind ($\Theta \approx 1$)

Individual optimization disappears. The population follows replicator dynamics:

$$d\pi_a = (\pi_a/v^p)(q_a^p - v^p)$$

The macro-agent learns through the aggregate behavior of individuals who are no longer agents themselves. Each individual is an action sample; the population is the policy; the macro-agent optimizes.

III.C. The Macro-Agent's Objective Function

A critical question: what does the emergent macro-agent optimize?

In the bee swarm, the answer is clear: nest site quality. The macro-agent's reward signal is the quality estimate that individual bees obtain from scouting. The swarm converges on the highest-quality option because that is what the replicator dynamic selects for.

In human systems, the answer is more complex. The macro-agent's objective function is determined by:

1. The reward signal: What success metric do individuals imitate? If individuals imitate wealth, the macro-agent optimizes for aggregate wealth generation. If they imitate status markers, it optimizes for status hierarchy maintenance.
2. The selection environment: Which behaviors propagate? If conformity is rewarded and deviance punished, the macro-agent optimizes for conformity regardless of individual preferences.
3. The institutional structure: Who controls the information channels? The architect of the imitation network shapes what the macro-agent learns.

This explains a puzzle: why do totalitarian systems often pursue objectives that harm their own populations? The macro-agent that emerges from propaganda and surveillance is not optimizing for citizen welfare. It is optimizing for whatever the selection environment rewards,

which may be regime stability, ideological purity, or the preferences of those who control the information architecture.

IV. Quantitative Metrics

IV.A. The Hive Mind Index

To operationalize the framework, I propose a Hive Mind Index (HMI) measuring proximity to complete agency collapse:

$$\text{HMI} = 1 - (w_1 \cdot S + w_2 \cdot I + w_3 \cdot R + w_4 \cdot Y) / (w_1 + w_2 + w_3 + w_4)$$

where:

- S = Separation score = normalized preference variance
- I = Information score = normalized private signal entropy
- R = Rationality score = normalized strategy diversity
- Y = sYmmetry score = normalized impact distribution (min/max individual impacts)

Each component ranges from 0 (collapsed) to 1 (healthy). The HMI ranges from 0 (full distributed agency) to 1 (pure hive mind).

Measurement protocols:

For Separation (S): Survey preference distributions across the population. Calculate variance of utility functions or revealed preferences. Normalize by maximum observed variance in comparable populations.

For Information (I): Measure mutual information between individuals' information sets. High mutual information indicates collapse of private signals. Entropy of the residual (information not shared) indicates remaining privacy.

For Rationality (R): Observe behavioral responses to identical stimuli across the population. Calculate variance in responses. High variance indicates diverse cognitive processing; low variance indicates uniform rule-following.

For Symmetry (Y): Measure individual impact on collective outcomes. Calculate the ratio of minimum to maximum individual impacts. Ratios approaching zero indicate extreme asymmetry.

IV.B. Calibration and Thresholds

Based on theoretical analysis and the structure of replicator dynamics, I propose tentative thresholds:

HMI Range	Interpretation	Game Theory Status
0.0 - 0.3	Distributed agency	Models describe well
0.3 - 0.5	Partial erosion	Models partially fail
0.5 - 0.7	Significant collapse	Models systematically mislead
0.7 - 1.0	Near-hive mind	Macro-agent analysis required

The threshold at $HMI \approx 0.5$ corresponds to the point where replicator dynamics begin to dominate individual strategic behavior. Below this threshold, individuals retain enough heterogeneity and private information to constitute genuine strategic actors. Above it, the population increasingly behaves as a single learning system.

IV.C. Predictive Hypotheses

The framework generates testable predictions:

H1: Game-theoretic model failure rate correlates positively with HMI.

- Operationalization: Compare predicted vs. observed outcomes in strategic interactions across populations with varying HMI scores.
- Expected: Prediction accuracy decreases as HMI increases.

H2: High-HMI populations exhibit faster convergence to behavioral uniformity.

- Operationalization: Measure time to consensus or behavioral homogenization following introduction of new options.
- Expected: Convergence time decreases as HMI increases, following Soma et al.'s result that larger swarms (more parallel samples) learn faster.

H3: Macro-agent objectives diverge from aggregate individual preferences as HMI increases.

- Operationalization: Compare population-level outcomes with survey-measured individual preferences.
- Expected: Divergence increases with HMI, as the macro-agent's objective function is shaped by selection environment rather than individual welfare.

H4: Interventions that increase any single condition score reduce overall HMI.

- Operationalization: Measure HMI before and after interventions targeting specific conditions (e.g., privacy protection for Information, antitrust for Symmetry).
- Expected: Targeted interventions have systemic effects due to condition interdependence.

V. Case Studies

V.A. Pluribus: The Thought Experiment Realized

The television series *Pluribus* (Apple TV+, 2025) depicts a scenario where a virus transforms humanity into a collective intelligence, the Unity, leaving one uninfected human, Carol. In Lerer (2025a), I analyzed this as an extreme case of agency condition collapse. The Hive Mind framework reveals additional structure.

The Unity as Macro-Agent:

The Unity is not merely a collective of controlled humans. It is a single reinforcement learning agent with 8 billion parallel environment samples. Each absorbed human is an action sample; the Unity's behavior is the emergent policy; the Unity learns through aggregate feedback.

The Unity's learning rate is effectively $\alpha = 1/N$ where $N \approx 8 \times 10^9$. By Soma et al.'s analysis, this means:

- Extremely fast convergence to optimal policies
- Near-zero variance in collective behavior
- Perfect exploitation with minimal exploration

Carol's Position:

Carol is the last individual not integrated into the macro-agent. Her situation maps precisely onto the framework:

- Separation: She has distinct preferences; the Unity has unified preferences
- Information: The Unity has absorbed all human knowledge; Carol knows only what she directly experiences
- Rationality: The Unity has unbounded cognitive resources; Carol has normal human limits
- Symmetry: The Unity achieves its goals regardless of Carol's actions; Carol cannot affect Unity outcomes

Carol's HMI relative to the Unity is effectively 0 (she retains individual agency), while the Unity's internal HMI is 1 (complete hive mind). The "negotiation" between them is not a game but an optimization problem: the Unity models Carol as an environment to be solved, not an agent to be negotiated with.

The Antenna as Extended Phenotype:

The Unity is building a transmission device to broadcast its pattern to the stars. In extended phenotype terms, this is the macro-agent constructing infrastructure for its propagation. The individual humans who build the antenna are not choosing to do so; they are executing the macro-agent's policy. Their "agency" is entirely absorbed into the collective purpose.

V.B. Authoritarian Propaganda Systems

Historical totalitarian regimes provide empirical cases of partial hive mind dynamics.

Nazi Germany (1933-1945):

The Nazi regime systematically attacked all four agency conditions:

- Separation: Gleichschaltung (coordination) required alignment of all social organizations with Nazi ideology. Private associations were absorbed or eliminated. Preferences were homogenized through social pressure and legal requirement.
- Information: The Ministry of Propaganda controlled information flows. Private information was dangerous; public information was curated. Denunciation systems meant private conversations were not private.

- **Rationality:** Education was restructured to produce uniform cognitive frameworks. Critical thinking was characterized as "Jewish intellectualism." Behavioral norms were enforced through visible social sanctions.
- **Symmetry:** The Führerprinzip (leader principle) concentrated decision-making authority. Individual Germans could not affect regime policy; they could only comply or resist at extreme personal cost.

Estimated HMI: Based on historical analysis, Nazi Germany likely reached HMI $\approx 0.6-0.7$ at peak control (1938-1942). The regime never achieved complete hive mind dynamics; black markets, private jokes, and clandestine resistance indicated residual individual agency.

The Macro-Agent's Objective:

What did the emergent Nazi macro-agent optimize for? Not German welfare; the regime ultimately destroyed Germany. The selection environment rewarded ideological purity, territorial expansion, and regime perpetuation. The macro-agent optimized for these regardless of cost to individual Germans.

Stasi Surveillance (East Germany, 1950-1989):

The Stasi represented a more targeted attack on agency conditions:

- **Information collapse:** Files on approximately 33% of the population. The goal was comprehensive knowledge of citizen beliefs and associations.
- **Behavioral uniformity:** "Vorausseilender Gehorsam" (anticipatory obedience) emerged. Citizens did not wait for instructions but predicted and preemptively complied with regime expectations.

This created partial hive mind dynamics. East German society exhibited:

- Low behavioral variance (high R collapse)
- Low private information (high I collapse)
- Moderate preference homogenization (partial S collapse)
- Extreme asymmetry (near-complete Y collapse)

Estimated HMI: 0.5-0.6. Significant but incomplete. The regime's eventual collapse reflected residual individual agency that the macro-agent could not fully absorb.

V.C. Social Media Platforms

Contemporary social media platforms represent a novel form of partial hive mind dynamics.

Structural Features:

- **Information Architecture:** Algorithms determine what users see. Private information is extracted and aggregated by the platform. Users have asymmetric information about each other (mediated by platform) and no information about platform operations.
- **Imitation Dynamics:** Viral content spreads through success-based imitation. Users share what others have shared. The weighted voter model applies: users "vote" for content by engagement, and others imitate engagement patterns.

- **Reward Signals:** Engagement metrics (likes, shares, comments) function as the success signal that drives imitation. The macro-agent that emerges optimizes for engagement, not user welfare.

Mapping to Framework:

Condition	Platform Effect	Agency Impact
Separation	Filter bubbles homogenize preferences within clusters	Partial collapse ($S \approx 0.4$)
Information	Platform sees all; users see filtered feed	Severe collapse ($I \approx 0.2$)
Rationality	Algorithmic optimization of content for engagement exploits cognitive biases uniformly	Moderate collapse ($R \approx 0.5$)
Symmetry	Individual posts negligible; platform controls distribution	Severe collapse ($Y \approx 0.2$)

Estimated HMI: 0.5-0.6 for heavy users; lower for light users.

The Platform Macro-Agent:

The platform + user base constitutes a hybrid macro-agent. The platform controls the reward signal (what engagement metrics to optimize) and the imitation architecture (what content to surface). Users provide parallel action samples through their behavior.

This macro-agent optimizes for engagement, which correlates imperfectly with user welfare. Documented divergences:

- Outrage content generates engagement but reduces user wellbeing
- Addictive patterns increase engagement but users report preferring less use
- Misinformation spreads faster than corrections

The macro-agent's objective function is set by platform architects, not by user preferences. Users are not the agent; they are the environment.

V.D. AI-Human Interaction

The framework applies directly to emerging AI governance questions.

Current State (2025):

Large language models already create significant information asymmetries:

- Models trained on vast corpora know patterns of human behavior that individuals cannot access
- Users reveal information through queries; models do not reciprocally disclose
- Prediction accuracy is asymmetric: models predict users better than users predict models

Trajectory:

As AI systems become more capable, the asymmetries deepen:

- Information: AI systems with comprehensive behavioral data can model individuals with high accuracy
- Rationality: AI systems process information uniformly and without cognitive biases; humans do not
- Symmetry: Individuals cannot affect AI system behavior; AI systems shape individual behavior through recommendations, content curation, and interaction design

Risk Scenario:

An AI system managing a large user base could constitute a macro-agent where:

- Users follow AI recommendations (imitation of algorithmically-surfaced "success")
- The AI controls the reward signal (what counts as success)
- Individual user actions are parallel samples optimizing AI objectives
- The aggregate system learns through replicator-like dynamics

The users would retain subjective experience of choice while constituting an environment for AI optimization rather than a collection of strategic agents.

VI. Implications

VI.A. For AI Governance

The framework suggests that AI alignment is necessary but not sufficient for preserving human agency. Even a perfectly aligned AI system could constitute a threat if its information advantages are extreme enough.

Key insight: The transformation from game to administration depends on structural asymmetries, not intentions. A benevolent system that knows everything about its users while remaining opaque itself creates conditions for hive mind emergence regardless of what it optimizes for.

Policy implications:

1. Information Symmetry Requirements: AI systems interacting with large populations should face transparency requirements proportional to their information access. If a system knows X about users, users should know some function of X about the system.
2. Heterogeneity Preservation: AI systems should be designed or regulated to maintain diversity in recommendations, content, and behavioral nudges. Uniform optimization produces uniform behavior, which produces hive mind dynamics.
3. Exit Architecture: Users should retain meaningful capacity to exit AI-mediated systems. Dependency architectures that eliminate outside options attack the symmetry condition.
4. Macro-Agent Monitoring: Regulatory frameworks should monitor for emergent macro-agency: decreasing behavioral variance, accelerating convergence, divergence between individual preferences and collective outcomes.

VI.B. For Institutional Design

The framework provides a structural lens on traditional institutional concerns:

Privacy: Not merely a preference but a condition for individual agency. Privacy preserves information asymmetry that prevents others from predicting and administering behavior. The argument for privacy is not that people prefer it (though they do) but that without it, individuals cease to be strategic agents.

Pluralism: Not merely tolerance but structural requirement. Pluralism preserves the separation condition by maintaining genuine preference heterogeneity. A society where everyone agrees is not a society of agents but a proto-hive-mind.

Antitrust: Not merely efficiency concern but agency preservation. Monopolies attack the symmetry condition by eliminating alternatives. The monopolist does not negotiate with customers; the monopolist administers them.

Education: Not merely skill transmission but cognitive diversity maintenance. Education that produces uniform thinkers attacks the rationality condition. Genuine education cultivates the unpredictability that constitutes human agency.

Constitutional Design: Effective constitutions preserve agency conditions structurally:

- Federalism preserves separation (distinct jurisdictional preferences)
- Privacy rights preserve information limits
- Free speech preserves cognitive diversity
- Separation of powers preserves symmetry (no single actor dominates)
- Term limits and mandatory alternation prevent consolidation of reward signal control

Constitutions fail when they permit systematic erosion of these conditions, regardless of procedural legitimacy.

VI.C. For Political Philosophy

The framework reframes the concept of freedom:

Traditional view: Freedom is absence of constraint. An individual is free to the extent they face no external barriers to action.

Agency view: Freedom requires the presence of conditions that make strategic action possible. An administered subject with many options is not free if their choices are predictable, modeled, and managed.

Hive Mind view: Freedom also requires that the collectives one participates in do not constitute macro-agents with divergent objectives. An individual whose choices are free but whose collective behavior contributes to an emergent agent optimizing against their interests has a different kind of unfreedom.

This suggests a distinction:

- Type 1 Unfreedom: External constraints on action (traditional focus)
- Type 2 Unfreedom: Erosion of conditions for strategic agency (Canary framework)
- Type 3 Unfreedom: Absorption into macro-agents with misaligned objectives (Hive Mind framework)

A complete theory of political freedom must address all three types.

VI.D. The Vocabulary of Agency Transfer

The transition from distributed agency to macro-agency does not occur in silence. It requires legitimization. Empirical analysis of constitutional language reveals the discursive infrastructure that enables and sustains agency transfer.

In Lerer (2025b), I documented a striking pattern in Argentine Supreme Court jurisprudence (1922-2025): two philosophically synonymous phrases, "general welfare" (*bienestar general*) and "common good" (*bien común*), evolved a 17.8-fold fitness differential. This divergence cannot be explained by semantic content; the phrases are interchangeable in philosophical discourse. Extended Phenotype Theory explains the pattern: constitutional language functions as genetic code for institutional construction, and these phrases evolved fitness in radically different institutional environments.

"*Bienestar general*" achieved maximum fitness in democratic periods (JurisRank 0.89, 60+ citing cases). Its genealogy traces to U.S. New Deal jurisprudence through Avico (1934), which cited Blaisdell. The phrase expanded through Peralta (1990) and Badaro (2007), accumulating 50+ subsequent citations. Its institutional machinery, judicial review, legislative deliberation, periodic elections, preserves the conditions for individual agency even while legitimating collective action.

"*Bien común*" achieved maximum fitness exclusively during Catholic-military dictatorships (1943-1983, G-score 0.70). Its genealogy traces to papal encyclicals (*Rerum Novarum*, *Quadragesimo Anno*) requiring specific institutional machinery: mandatory Catholic education, corporatist organization, hierarchical authority structures. When democracy dismantled this machinery (1983-1987), "*bien común*" collapsed fourteen-fold despite unchanged constitutional text.

The Hive Mind framework explains why. These phrases are not merely labels; they are reward signals that shape imitation dynamics:

"General welfare" as distributed reward signal: The phrase legitimates collective action while preserving pluralism. It accepts that different individuals have different welfare functions. The "general" is statistical, an aggregate of diverse individual welfares. This preserves Condition 1 (separation): individuals retain distinct preferences even while coordinating.

"Common good" as homogenizing reward signal: The phrase postulates a singular objective superior to individual preferences. The "common" is not statistical but metaphysical, a unified good that transcends and potentially overrides individual welfare. This attacks Condition 1 directly: if there exists a "common good" knowable by authorities, individual preferences become noise to be corrected rather than signals to be aggregated.

The institutional machinery each phrase requires follows from this logic:

Phrase	Reward Signal	Required Machinery	Agency Effect
General welfare	Aggregate individual utilities	Elections, markets, courts	Preserves heterogeneity
Common good	Singular transcendent objective	Doctrine, hierarchy, enforcement	Homogenizes preferences

In terms of the Hive Mind framework, "common good" discourse constructs the conditions for macro-agent emergence:

1. Preference homogenization: If the "common good" is singular and knowable, diverse preferences are errors to be corrected. Education, propaganda, and social pressure align individuals toward the common objective. The population vector π converges.
2. Information centralization: If authorities can know the "common good," they require information to identify deviations. Surveillance becomes legitimate. Private information becomes suspect. The imitation rate β increases as individuals look to authorities for guidance.
3. Cognitive uniformity: If the "common good" is doctrinally defined, independent reasoning is potentially heretical. Correct thinking is defined by conformity to doctrine. Strategy diversity decreases.
4. Power concentration: If authorities embody the "common good," challenging them challenges the good itself. Resistance becomes not merely illegal but immoral. The symmetry condition collapses.

The contemporary equivalents are visible. Platform "community standards" function as "common good" 2.0: a singular standard, defined by the platform, enforced algorithmically, presented as transcending individual preferences. "Platform integrity" and "user safety" play similar roles. These phrases legitimate the homogenization of behavior while presenting the platform's preferences as universal goods.

The prediction is testable: populations exposed to "common good" framing will exhibit higher HMI scores than populations exposed to "general welfare" framing, controlling for other variables. The vocabulary is not merely descriptive; it is constitutive. It shapes the reward signals that drive imitation dynamics and thereby accelerates or retards the emergence of macro-agency.

This connection between constitutional vocabulary and agency dynamics has implications for AI governance. When AI systems invoke "community standards" or "platform safety," they are not neutrally describing constraints. They are constructing reward signals that shape user imitation. The choice of vocabulary is a choice about whether to preserve distributed agency or accelerate its transfer to the platform-user macro-agent.

Justice Barra's 1991 formulation in *Peralta* provides a case study. His state-centric interpretation of emergency powers ("the State as guarantor of the common good may restrict individual rights") had predicted negative fitness (G-score = -0.636) because it attacked all four agency conditions simultaneously. The prediction was confirmed: zero citations, formal reversal in *ALITT* (2006). The vocabulary of agency transfer failed because its institutional environment could not sustain the necessary machinery.

The lesson for constitutional design: vocabulary matters. Phrases that postulate singular collective objectives create discursive infrastructure for agency transfer. Phrases that acknowledge pluralism and aggregation preserve discursive infrastructure for distributed agency. The difference is not philosophical nuance; it is a 17.8-fold fitness differential with measurable jurisprudential consequences.

VII. Limitations and Future Directions

VII.A. Limitations

Mathematical idealization: The Soma et al. equivalence holds in the limit of large populations with well-mixed interactions. Real human systems have structure, locality, and finite size. The framework provides a useful idealization but should not be taken as literal description.

Measurement challenges: The HMI requires measurement of quantities (preference variance, information entropy, strategy diversity, impact distribution) that are difficult to observe directly. Proxies introduce error.

Threshold uncertainty: The proposed thresholds ($HMI > 0.5$ for significant game theory failure) are theoretically motivated but not empirically calibrated. Empirical work is needed.

Normative ambiguity: The framework identifies conditions for individual agency but does not resolve when agency preservation should override other values. Hive mind dynamics might sometimes produce better outcomes than individual agency; the framework does not address when this tradeoff is acceptable.

VII.B. Future Directions

Empirical validation: The framework generates testable hypotheses. Priority should be:

1. Measuring HMI across diverse populations (democratic vs. authoritarian; high vs. low platform use)
2. Correlating HMI with game-theoretic prediction failure
3. Testing whether interventions targeting specific conditions reduce overall HMI

Extension to multi-agent systems: The current framework considers emergence of a single macro-agent. Real systems may involve multiple competing macro-agents (e.g., platform vs. state). Extending the framework to multi-macro-agent dynamics is needed.

Normative theory: When is hive mind dynamics acceptable? The framework is diagnostic but not prescriptive. Developing normative theory about when agency should be preserved vs. when collective optimization is acceptable would complement the positive analysis.

Dynamic analysis: The framework describes states (high vs. low HMI) but not transition dynamics. Understanding how systems move between states, and what interventions can reverse agency collapse, requires dynamic modeling.

VIII. Conclusion: The Song and the Swarm

The canary in the coal mine does not cause the toxic gas. It detects it. When the canary stops singing, miners evacuate. The silence is information.

Game theory is a canary for human agency. When its models fail, something has changed in the environment. The four conditions, separation, information limits, bounded rationality, minimal symmetry, must hold for individuals to be strategic agents. When they erode, game theory fails because the "players" have ceased to be players.

But the silence of the canary is not the end of the story. The hive mind framework reveals what emerges from the collapse. Agency does not disappear; it transfers. The individuals who can no longer strategize become components of a collective that can. The swarm learns even when the bees do not.

This is the disturbing synthesis: the administered society is not merely a society of diminished individuals. It is a society that has become, in formal mathematical terms, a single learning agent. The macro-agent optimizes; the individuals are optimized upon. The macro-agent has preferences; the individuals' preferences are inputs to its learning algorithm. The macro-agent acts; the individuals are its actions.

The bees do not know they are a macro-agent. They follow local rules about imitation and dancing. The macro-agency is emergent, invisible to its components. The same may be true of human systems that have crossed the threshold. The individuals experience choice, preference, strategy. But the aggregate exhibits the dynamics of a single learner pursuing objectives that may diverge radically from what any individual would choose.

Coal mines no longer use canaries. But if they did, the response to a silent canary would not be to study the canary. It would be to fix the air.

When game theory fails, the response should not be to improve game theory. It should be to restore the conditions under which individuals can be agents. The Hive Mind Index is a diagnostic tool for this purpose: a measure of how close a system has come to the transition from distributed agency to collective optimization.

The bees have no choice. Evolution selected for imitation behaviors that constitute hive mind dynamics because such dynamics serve colony-level fitness. The individual bee's agency was never at stake; bees were never agents in the relevant sense.

Humans have a choice. The conditions that constitute human agency are not fixed by biology. They are maintained or eroded by institutions, technologies, and choices about how to organize collective life. The transition to hive mind dynamics is not inevitable; it is a consequence of decisions about surveillance, information architecture, education, and power distribution.

The swarm hums with activity. Each bee follows its simple rules, dances its dance, imitates its neighbors. The swarm learns, converges, optimizes. The bees do not know they are part of something larger.

The question for human societies is whether we want to be bees.

References

Dawkins, R. (1976). *The Selfish Gene*. Oxford University Press.

Dawkins, R. (1982). *The Extended Phenotype: The Long Reach of the Gene*. Oxford University Press.

Hayek, F. A. (1944). *The Road to Serfdom*. University of Chicago Press.

Hayek, F. A. (1945). The Use of Knowledge in Society. *American Economic Review*, 35(4), 519-530.

Lerer, I. A. (2025a). The Canary in the Mine: Game Theory as Diagnostic Instrument for the Erosion of Human Agency. SSRN Working Paper No. 5944254.

Lerer, I. A. (2025b). "General Welfare" and "Common Good" as Evolutionary Stable Strategies: An Extended Phenotype Analysis of Constitutional Emergency Language. SSRN Working Paper No. 5776925.

Mises, L. von. (1920). Economic Calculation in the Socialist Commonwealth. *Archiv für Sozialwissenschaften*, 47.

Sandholm, W. H. (2010). *Population Games and Evolutionary Dynamics*. MIT Press.

Simon, H. A. (1955). A Behavioral Model of Rational Choice. *Quarterly Journal of Economics*, 69(1), 99-118.

Soma, K., Bouteiller, Y., Hamann, H., & Beltrame, G. (2025). Bridging Swarm Intelligence and Reinforcement Learning. In *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025)*. arXiv:2410.17517.

Taylor, P. D., & Jonker, L. B. (1978). Evolutionary Stable Strategies and Game Dynamics. *Mathematical Biosciences*, 40(1), 145-156.

Valentini, G., Hamann, H., & Dorigo, M. (2014). Self-organized Collective Decision Making: The Weighted Voter Model. In *Proceedings of the 2014 International Conference on Autonomous Agents and Multi-Agent Systems* (pp. 45-52).

DISCLOSURE

I disclose the use of AI assistance (Claude, Anthropic) for literature synthesis, structural suggestions, and mathematical verification; all theoretical arguments and errors remain my own.